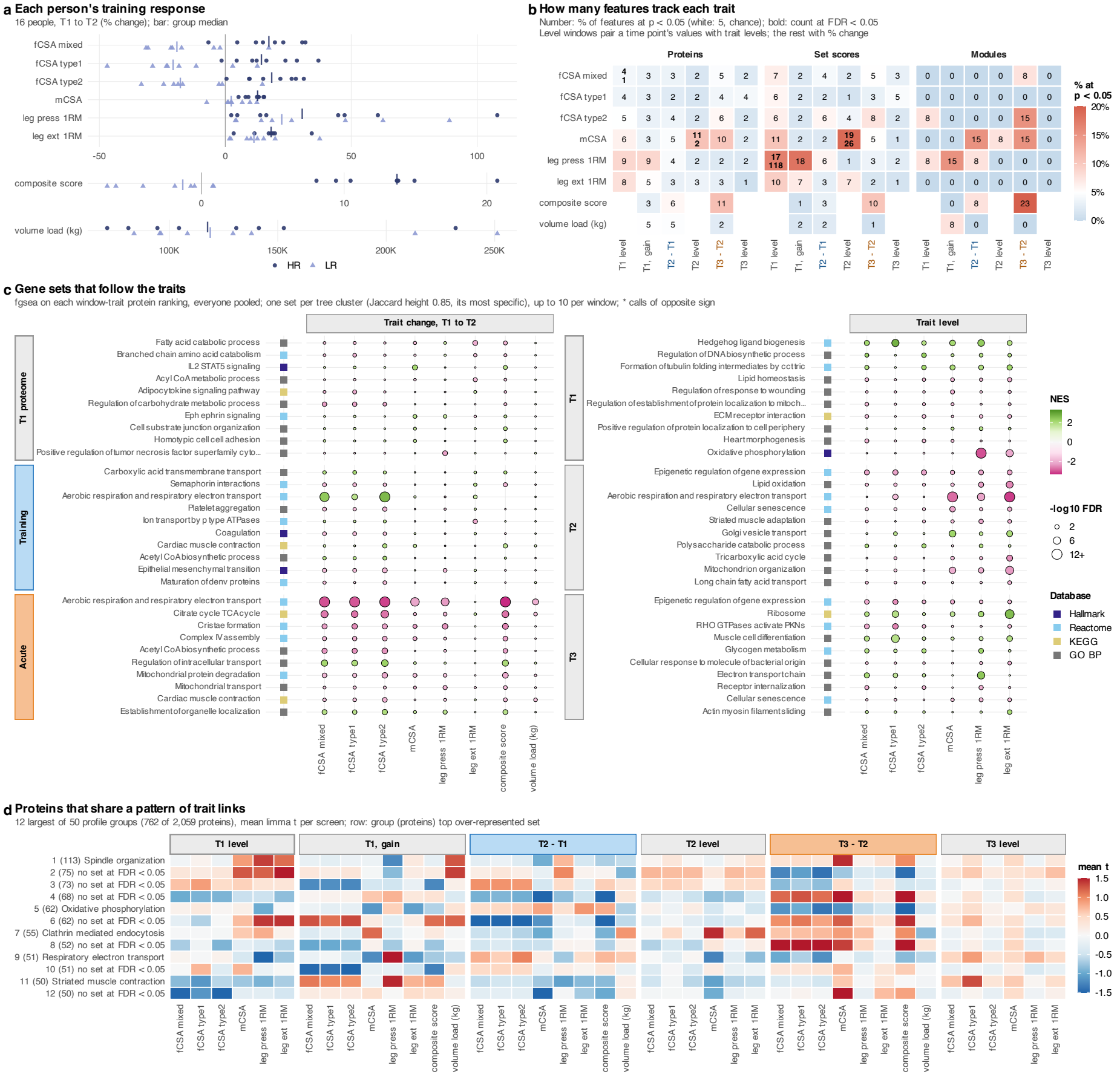

a Each person's training response
16 people, T1 to T2 (% change); bar: group median
fCSA mixed
fCSA type1
fCSA type2
mCSA
leg press 1RM
leg ext 1RM
composite score
volume load (kg)
HR
LR
b How many features track each trait
Number: % of features at p < 0.05 (white: 5, chance); bold: count at FDR < 0.05
Level windows pair a time point's values with trait levels; the rest with % change
Proteins
Set scores
Modules
% at p < 0.05
T1 level
T1, gain
T2 - T1
T2 level
T3 - T2
T3 level
c Gene sets that follow the traits
fgsea on each window-trait protein ranking, everyone pooled; one set per tree cluster (Jaccard height 0.85, its most specific), up to 10 per window; * calls of opposite sign
Trait change, T1 to T2
Trait level
T1 proteome
Training
Acute
T1
T2
T3
Fatty acid catabolic process
Branched chain amino acid catabolism
IL2 STAT5 signaling
Acyl CoA metabolic process
Adipocytokine signaling pathway
Regulation of carbohydrate metabolic process
Eph ephrin signaling
Cell substrate junction organization
Homotypic cell cell adhesion
Positive regulation of tumor necrosis factor superfamily cyto...
Carboxylic acid transmembrane transport
Semaphorin interactions
Aerobic respiration and respiratory electron transport
Platelet aggregation
Ion transport by p type ATPases
Coagulation
Cardiac muscle contraction
Acetyl CoA biosynthetic process
Epithelial mesenchymal transition
Maturation of denv proteins
Aerobic respiration and respiratory electron transport
Citrate cycle TCA cycle
Cristae formation
Complex IV assembly
Acetyl CoA biosynthetic process
Regulation of intracellular transport
Mitochondrial protein degradation
Mitochondrial transport
Cardiac muscle contraction
Establishment of organelle localization
Hedgehog ligand biogenesis
Regulation of DNA biosynthetic process
Formation of tubulin folding intermediates by cct tric
Lipid homeostasis
Regulation of response to wounding
Regulation of establishment of protein localization to mitoch...
ECM receptor interaction
Positive regulation of protein localization to cell periphery
Heart morphogenesis
Oxidative phosphorylation
Epigenetic regulation of gene expression
Lipid oxidation
Aerobic respiration and respiratory electron transport
Cellular senescence
Striated muscle adaptation
Golgi vesicle transport
Polysaccharide catabolic process
Tricarboxylic acid cycle
Mitochondrion organization
Long chain fatty acid transport
Epigenetic regulation of gene expression
Ribosome
RHO GTPases activate PKNs
Muscle cell differentiation
Glycogen metabolism
Cellular response to molecule of bacterial origin
Electron transport chain
Receptor internalization
Cellular senescence
Actin myosin filaments sliding
fCSA mixed
fCSA type1
fCSA type2
mCSA
leg press 1RM
leg ext 1RM
composite score
volume load (kg)
NES
-log10 FDR
Database
Hallmark
Reactome
KEGG
GO BP
d Proteins that share a pattern of trait links
12 largest of 50 profile groups (762 of 2,059 proteins), mean limma t per screen; row: group (proteins) top over-represented set
T1 level
T1, gain
T2 - T1
T2 level
T3 - T2
T3 level
1 (113) Spindle organization
2 (75) no set at FDR < 0.05
3 (73) no set at FDR < 0.05
4 (68) no set at FDR < 0.05
5 (62) Oxidative phosphorylation
6 (62) no set at FDR < 0.05
7 (55) Clathrin mediated endocytosis
8 (52) no set at FDR < 0.05
9 (51) Respiratory electron transport
10 (51) no set at FDR < 0.05
11 (50) Striated muscle contraction
12 (50) no set at FDR < 0.05
mean t